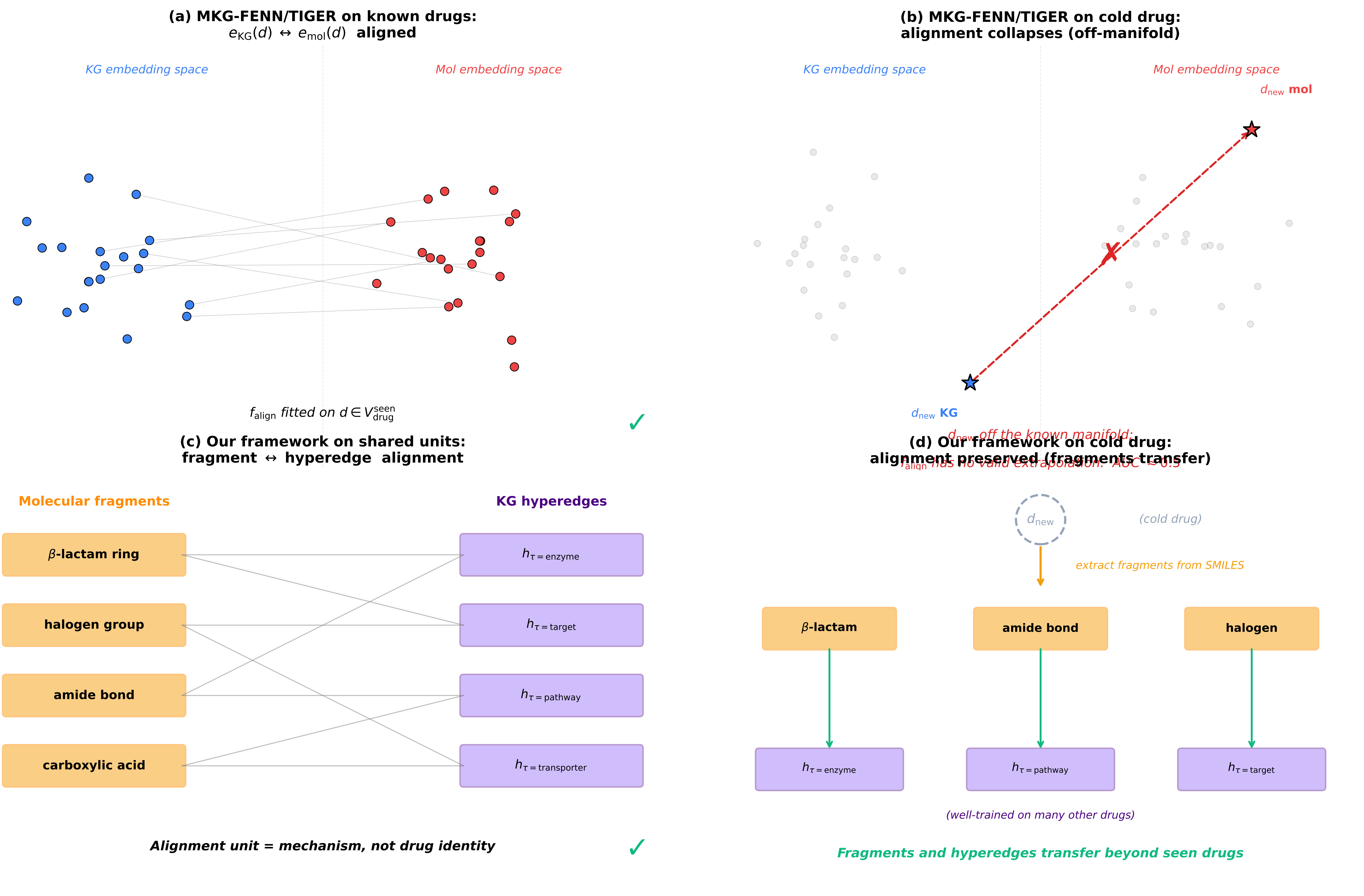
Figure 2. Multi-modal alignment object determines cold-start transferability.
Top: drug-identity alignment (MKG-FENN, TIGER) collapses on cold drug.
Bottom: mechanism alignment (ours) is preserved.



Multi-modal DDI methods like MKG-FENN and TIGER align KG and molecular views at the drug-identity level.
Under cold-start, the new drug's identity is OOD; the learned alignment manifold cannot extrapolate.
Our framework aligns at the mechanism level — molecular fragments map to hyperedge mechanism units —
and because fragments and mechanism units transfer beyond seen drugs, alignment survives cold-start.